

NETFLIX DATA ANALYSIS & EDA

Uncovering Content Strategy through Python

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CONTENT ANALYTICS: THE ENGINE OF STREAMING

- ▶ **Context:** In the modern streaming war, success isn't accidental; it's calculated.
- ▶ **The Goal:** Data doesn't just measure past success—it informs future architectural decisions.
- ▶ **Project Scope:** Leveraging historical data to decode the DNA of the world's largest content library.



THE CHALLENGE: DECIPHERING THE LIBRARY

Core Question: With thousands of titles added over a decade, what are the underlying strategic patterns?

- Identifying content saturation points.
- Uncovering geographic biases in acquisition.
- Tracing temporal trends hidden within the metadata.

ANALYTICAL OBJECTIVES



1. Analyze Distribution

Movies vs. TV Balance



2. Map Trends

Geography & Demographics



3. Decode Strategy

Genre & Timing

THE SOURCE CODE

Tech Spec

DATASET STATISTICS

8,790

Unique Records

10

Features

```
show_id  
type  
title  
director  
country  
date_added  
release_year  
rating  
duration  
listed_in
```


REFINING THE RAW DATA



1. Integrity Check

Confirmed 0 duplicate rows to ensure statistical validity.



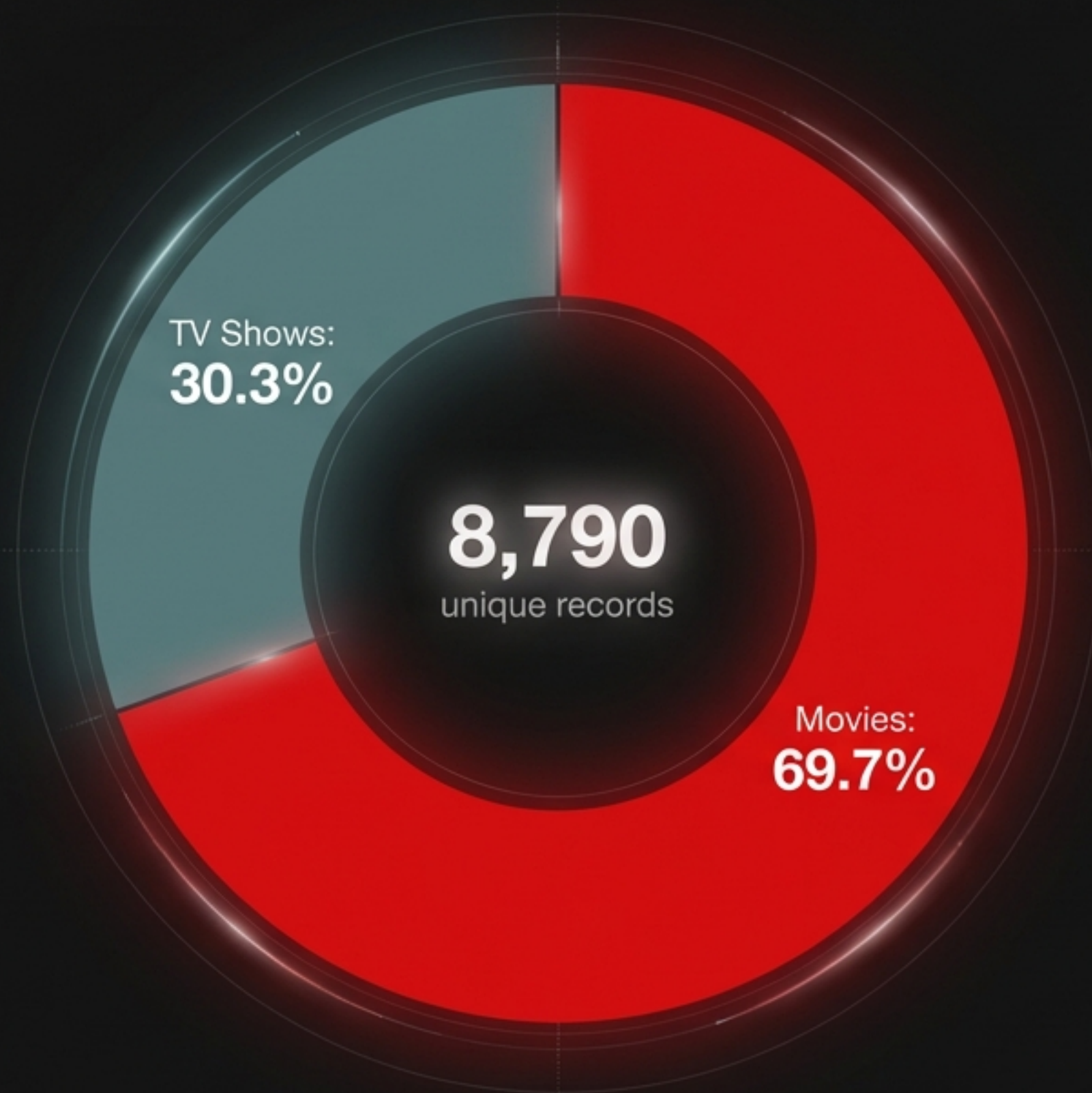
2. Handling Nulls

Managed missing values in Director, Cast, and Country columns.



3. Feature Engineering

Converted 'date_added' strings to DateTime objects for time-series analysis.



THE CINEMATIC EMPHASIS

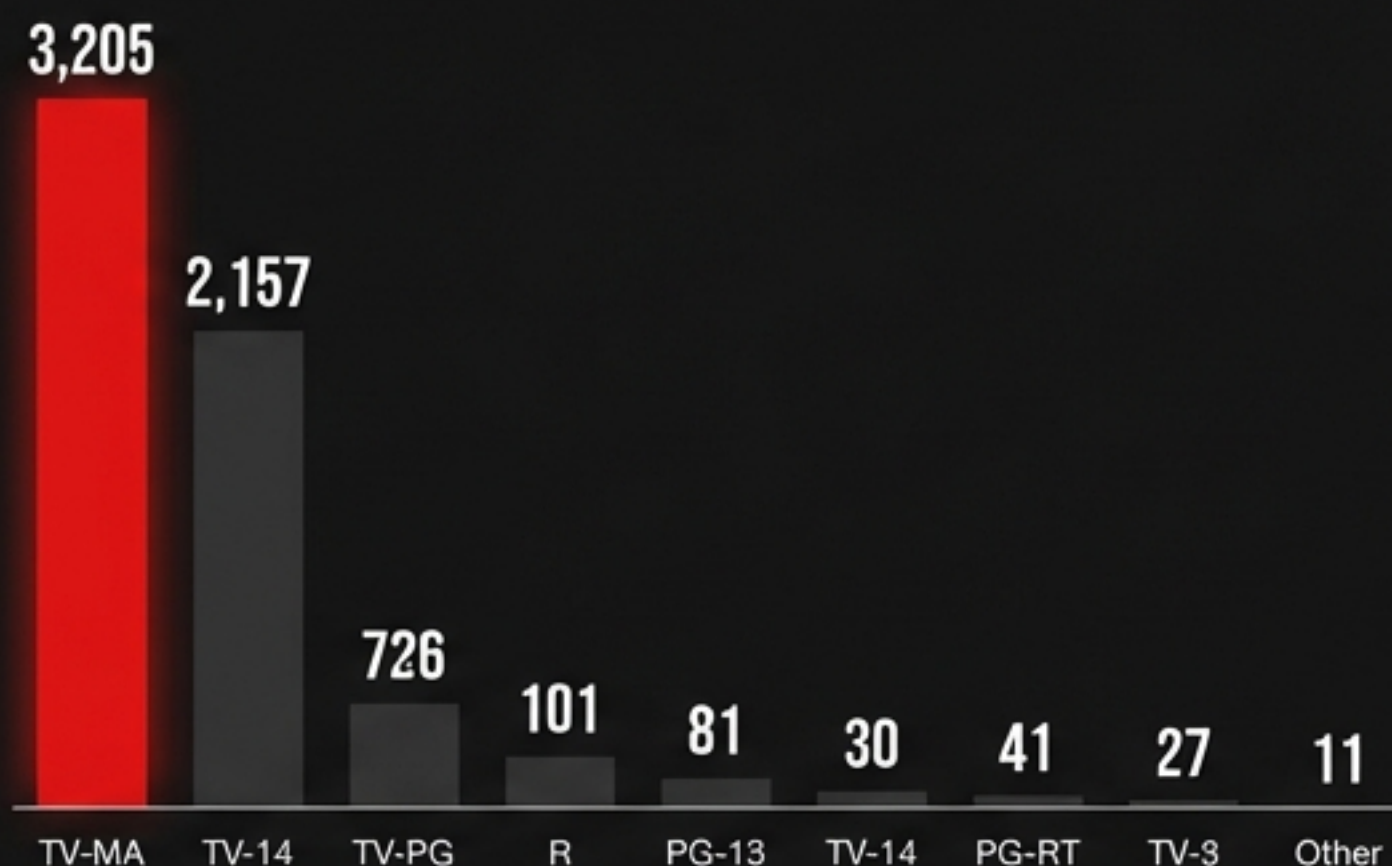
Despite the cultural buzz around binge-worthy series, the library's foundation is structurally built on a massive catalog of feature films (6,126 titles vs 2,664 TV shows).

GLOBAL REACH, MATURE AUDIENCE

TOP 3 CONTENT HUBS



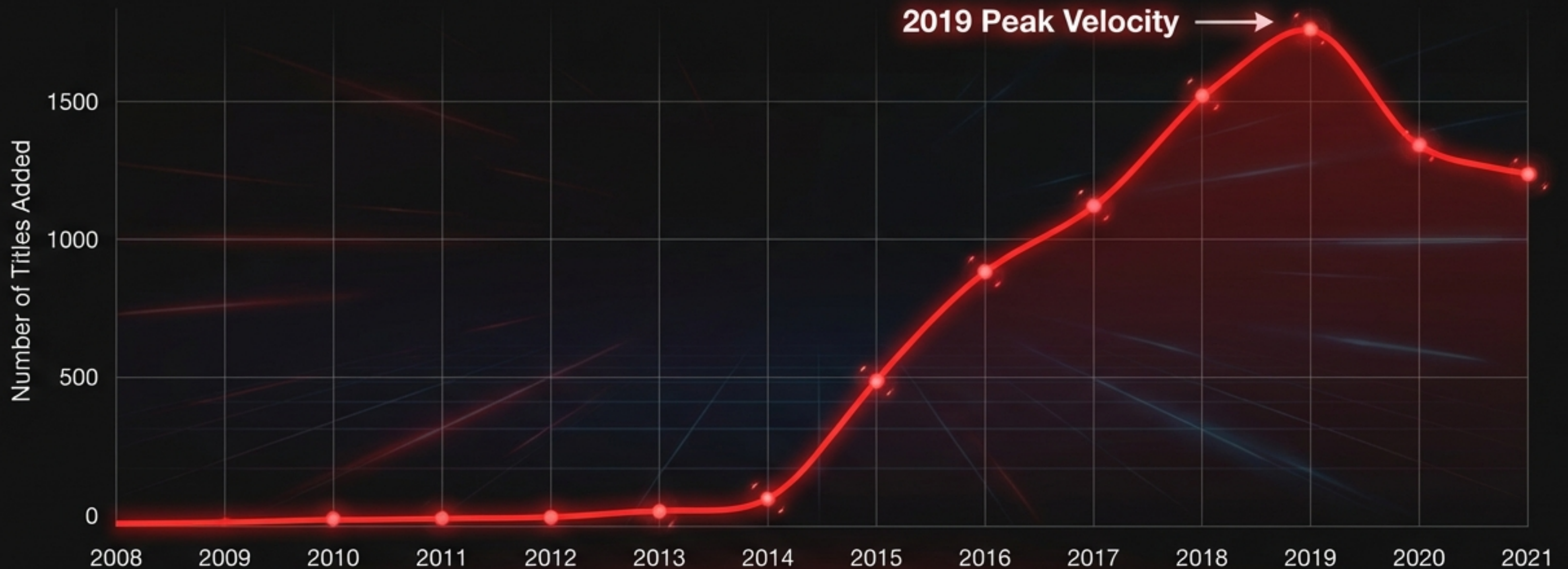
CONTENT RATINGS



The platform is heavily Western-focused but maintains India as a major hub.
Content is primarily targeted at adults (TV-MA).

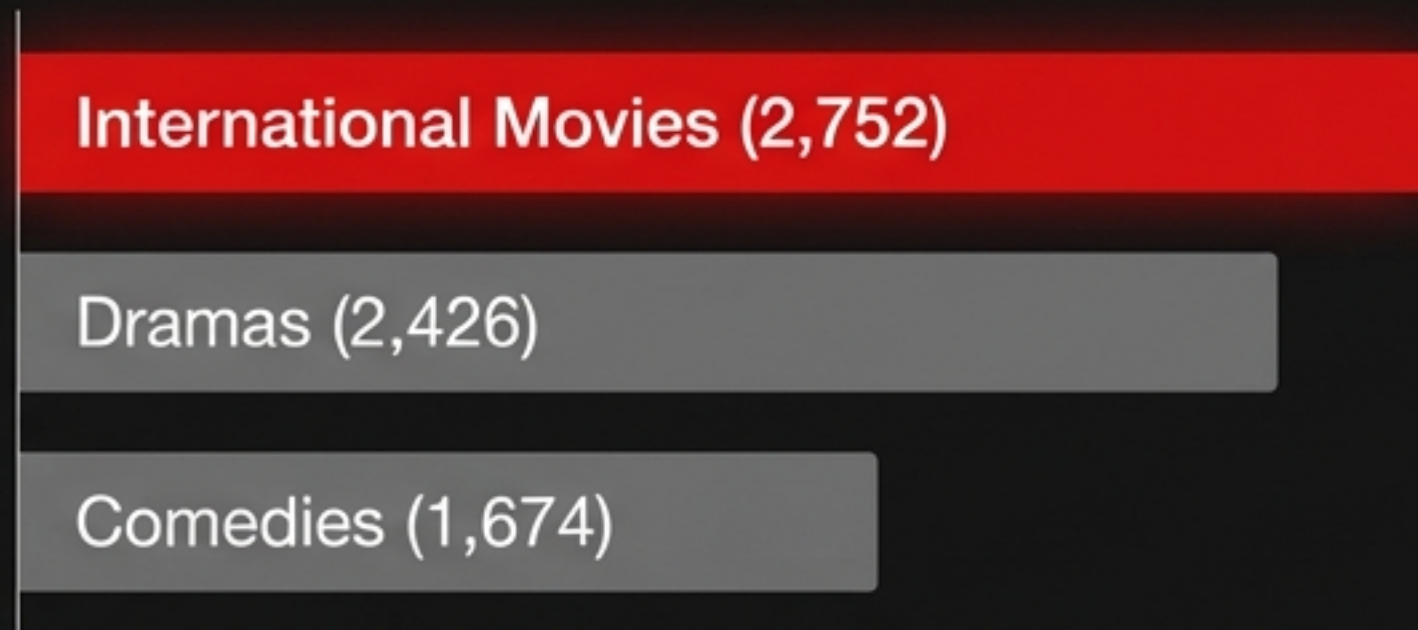
THE CONTENT EXPLOSION (2015-2019)

Rapid acceleration of content acquisition starting in 2015.

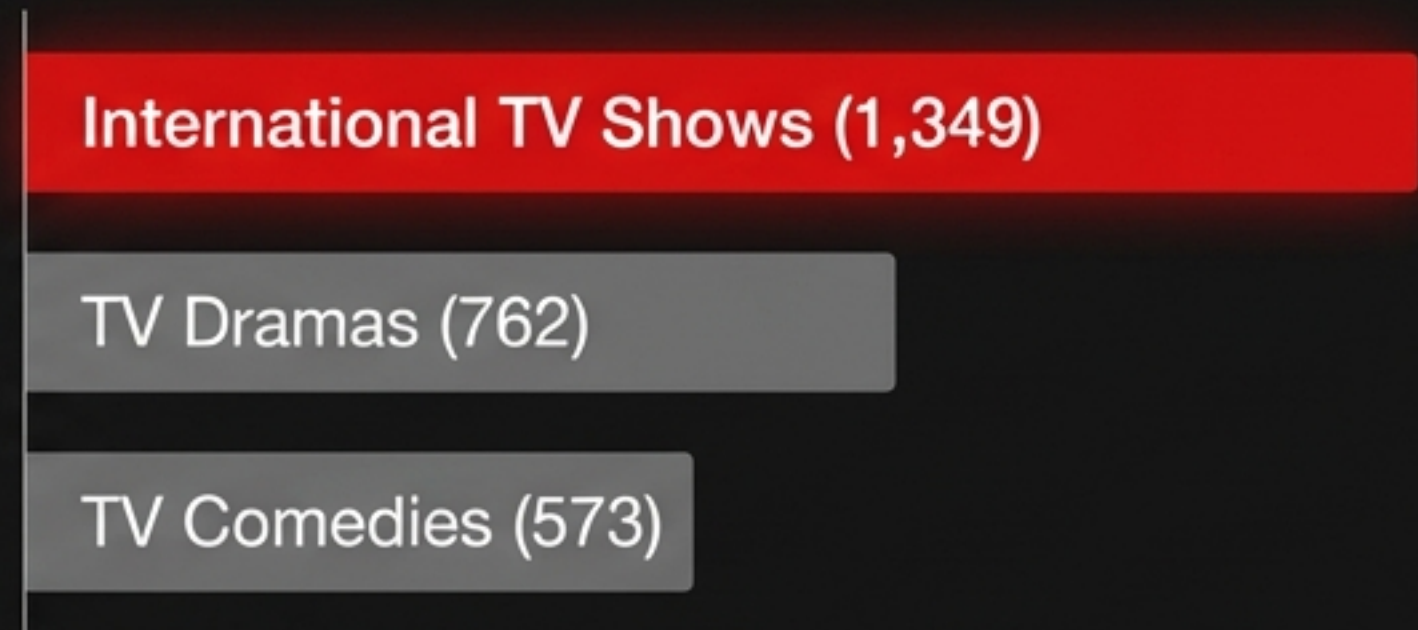


WHAT WE WATCH: GENRE DOMINANCE

TOP MOVIE GENRES



TOP TV GENRES



‘International’ is the leading category tag across both formats, reinforcing a strategy of global content importation.

STRATEGIC IMPLICATIONS



GO GLOBAL

The data proves a massive investment in International content.

Strategy: Continue leveraging local markets like India and the UK.



THE RETENTION FACTOR

While movies offer volume (69%), the rising trend in TV shows is crucial for long-term subscriber retention.



ADULT-FIRST

The dominance of TV-MA suggests the primary subscriber base is adults; acquisition should align with this demographic.

CONCLUSION & FUTURE SCOPE

Conclusion: We successfully mapped a library defined by International content, Mature ratings, and a 2019 peak volume.

Future Scope: Move from EDA to Machine Learning by implementing Recommendation Systems based on description NLP and Director Clustering.

THANK YOU

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Questions?